

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

CN Lecture 05: Routing (Routing Information Protocol - RIP)

OBE Based Questions and Answers

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1. A company has recently expanded its network by adding several branch offices. Explain why routing is necessary and describe how a router forwards packets to the correct destination.

Answer:

Routing is the process of forwarding packets from a source network to a destination network using a router.

A router examines its routing table to determine the best path. The routing table contains the destination network, next-hop address, metric, and outgoing interface. Based on this information, the router forwards the packet through the appropriate interface toward its destination.

2. A small office network contains only two routers with fixed network paths that rarely change. Recommend whether Static Routing or Dynamic Routing should be used and justify your answer.

Answer:

Static Routing should be used.

Reasons:

- Routes are configured manually.
- Suitable for small and stable networks.
- Simple to implement.
- Generates no routing update traffic.
- Requires minimal system resources.

Therefore, Static Routing is the most appropriate solution for this scenario.

3. A university campus network frequently changes because new departments are added every semester. Explain why Dynamic Routing is more suitable than Static Routing.

Answer:

Dynamic Routing automatically exchanges routing information among routers.

Its advantages include:

- Automatic route discovery.
- Automatic routing table updates.
- Adaptation to network topology changes.
- Automatic selection of the best path.
- Suitable for medium and large networks.

Hence, Dynamic Routing minimizes administrative effort and improves network scalability.

4. An Internet Service Provider manages several networks under one administrative authority. Identify the appropriate routing protocol category and explain your answer.

Answer:

The appropriate category is the **Interior Gateway Protocol (IGP)**.

IGP is used for routing within a single Autonomous System (AS), where all networks are managed by the same administrator.

Examples include:

- RIP
- EIGRP
- OSPF
- IS-IS

5. **Two different Internet Service Providers need to exchange routing information between their networks. Which routing protocol category should be used? Explain your answer.**

Answer:

The appropriate category is the **Exterior Gateway Protocol (EGP)**.

EGP is designed for communication between different Autonomous Systems.

The most widely used EGP is:

- Border Gateway Protocol (BGP)

BGP enables Internet Service Providers to exchange routing information across the Internet.

6. **A network engineer must choose between Distance Vector Routing and Link State Routing for a growing enterprise network. Compare these routing approaches.**

Answer:

Feature	Distance Vector	Link State
Information Exchange	Neighbor routers only	All routers
Network View	Partial	Complete topology
Updates	Periodic full updates	Partial updates
Algorithm	Bellman-Ford	Link-state algorithm
Example	RIP	OSPF

Link State Routing is generally more efficient for large networks because it maintains a complete network topology and sends updates only when changes occur.

7. **A router receives multiple routes to the same destination from different routing sources. Explain how Administrative Distance (AD) determines the preferred route.**

Answer:

Administrative Distance (AD) measures the trustworthiness of a routing source.

- Lower AD indicates a more preferred route.
- AD values range from 0 to 255.
- A directly connected network has an AD of 0, which is the highest preference.

The router installs the route with the lowest Administrative Distance into its routing table.

8. **A network administrator plans to deploy RIP in a campus network. Explain how RIP determines the best path and identify its limitations.**

Answer:

RIP uses **Hop Count** as its routing metric.

Characteristics:

- Lowest hop count is selected.
- Maximum allowable hop count is 15.
- Networks requiring more than 15 hops are considered unreachable.
- Routing updates are exchanged every 30 seconds.

Although RIP is simple to configure, it is not suitable for very large networks because of the hop-count limitation.

9. A company is upgrading from RIPv1 to RIPv2. Explain the improvements provided by RIPv2.

Answer:

RIPv2 offers several improvements over RIPv1:

- Supports classless addressing (VLSM).
- Uses multicast (224.0.0.9) instead of broadcast.
- Supports authentication.
- Exchanges only Version 2 routing updates.

These enhancements improve scalability, security, and routing efficiency.

10. A newly installed router initially knows only its directly connected networks. Explain how the RIP routing table becomes complete after startup.

Answer:

Initially, each router contains only directly connected routes.

The RIP routing process is:

- (a) Routers exchange routing updates with neighboring routers.
- (b) Newly learned routes are added to the routing table.
- (c) Hop counts are incremented.
- (d) Periodic updates continue every 30 seconds.
- (e) Eventually, every router learns all reachable networks.

This process is known as routing table convergence.

11. A network link suddenly fails between two routers. Explain how RIP updates the routing tables after detecting the failure.

Answer:

When a route becomes unavailable:

- RIP waits for route updates.
- If no update is received within 180 seconds, the route is marked invalid.
- The router removes the invalid route from its routing table.
- Neighboring routers are informed through routing updates.
- All routers update their routing tables to reflect the network change.

This process allows the network to converge to the new topology.

12. Evaluate the suitability of Static Routing, Dynamic Routing, and RIP for different networking environments.

Answer:

Technology	Suitable Environment
Static Routing	Small, stable networks with infrequent topology changes
Dynamic Routing	Medium and large networks requiring automatic route updates
RIP	Small to medium-sized networks with fewer than 16 hops

Static Routing is simple but requires manual configuration.

Dynamic Routing automatically adapts to network changes.

RIP is easy to configure but is limited by its hop-count metric and periodic update mechanism.